



Raspberry Pi Initialization

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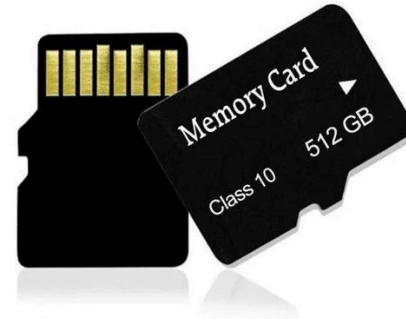


Pre-requisites

- Raspberry Pi – 4B
- SD card/ USB Drive
- Laptop
- SD card reader (optional)

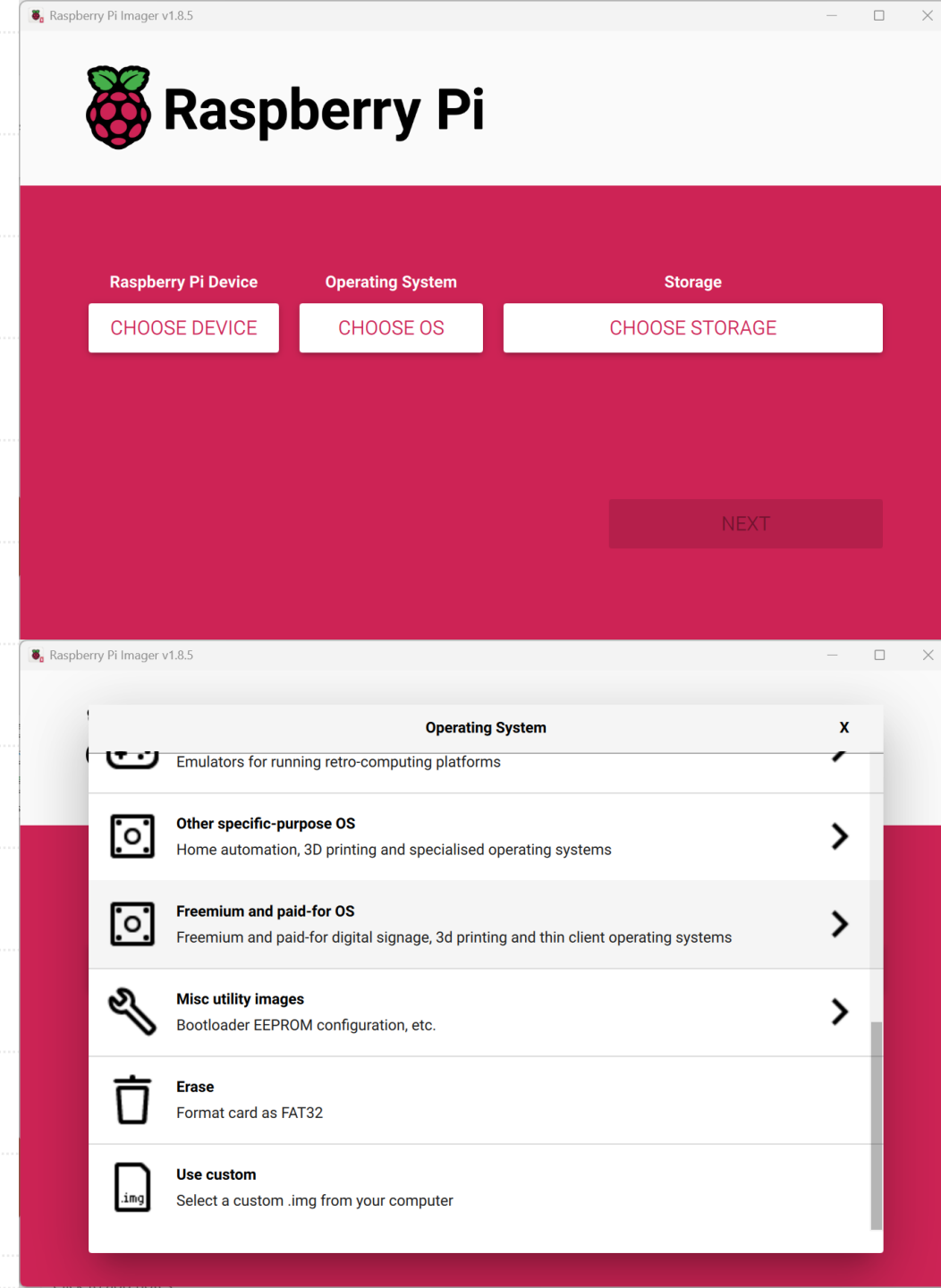
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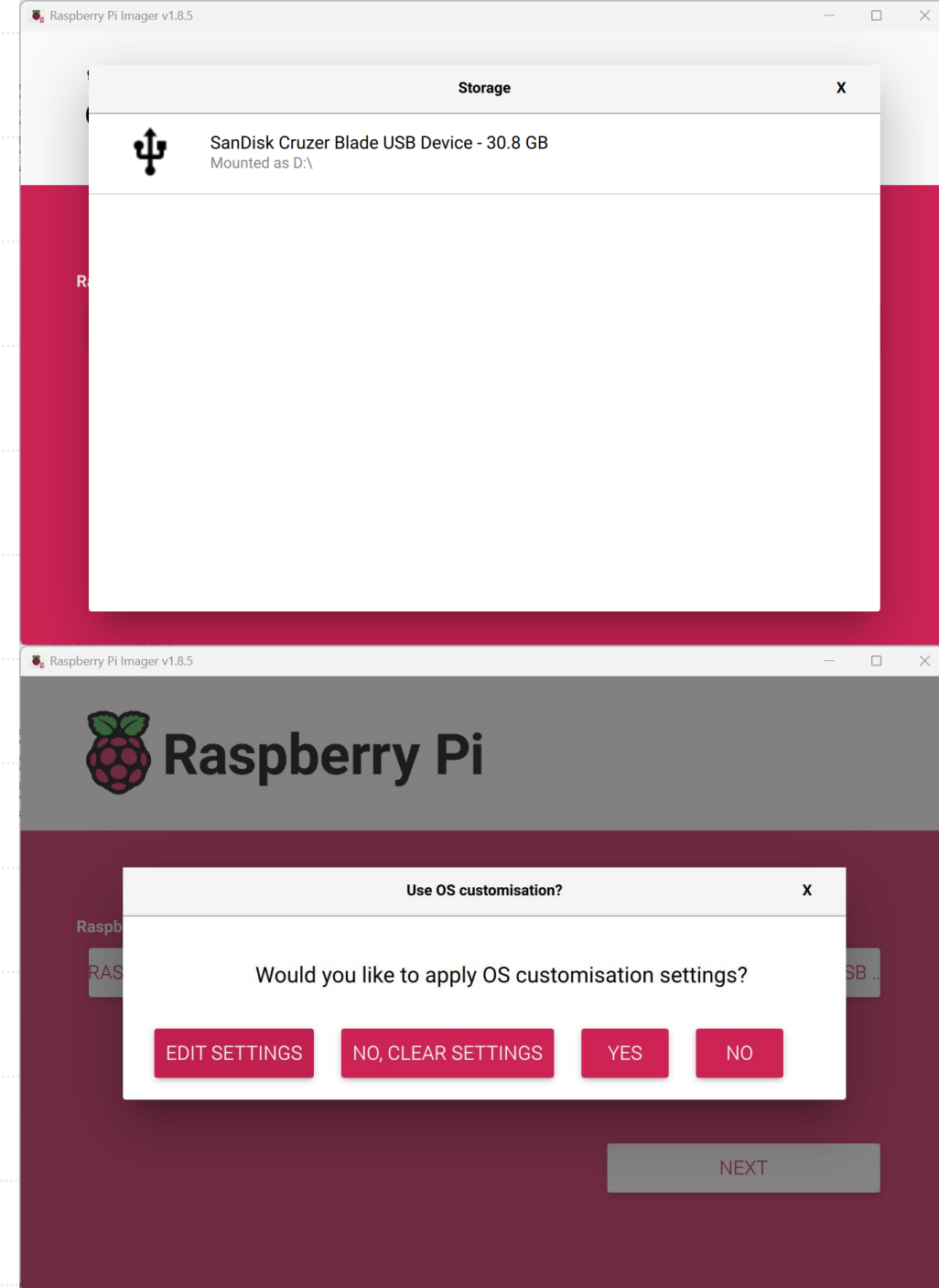
Raspberry Pi Imager

- Software to flash OS in Rpi
- Device = Raspberry Pi 4
- OS = Use Custom (Provided in Teams, img extension)



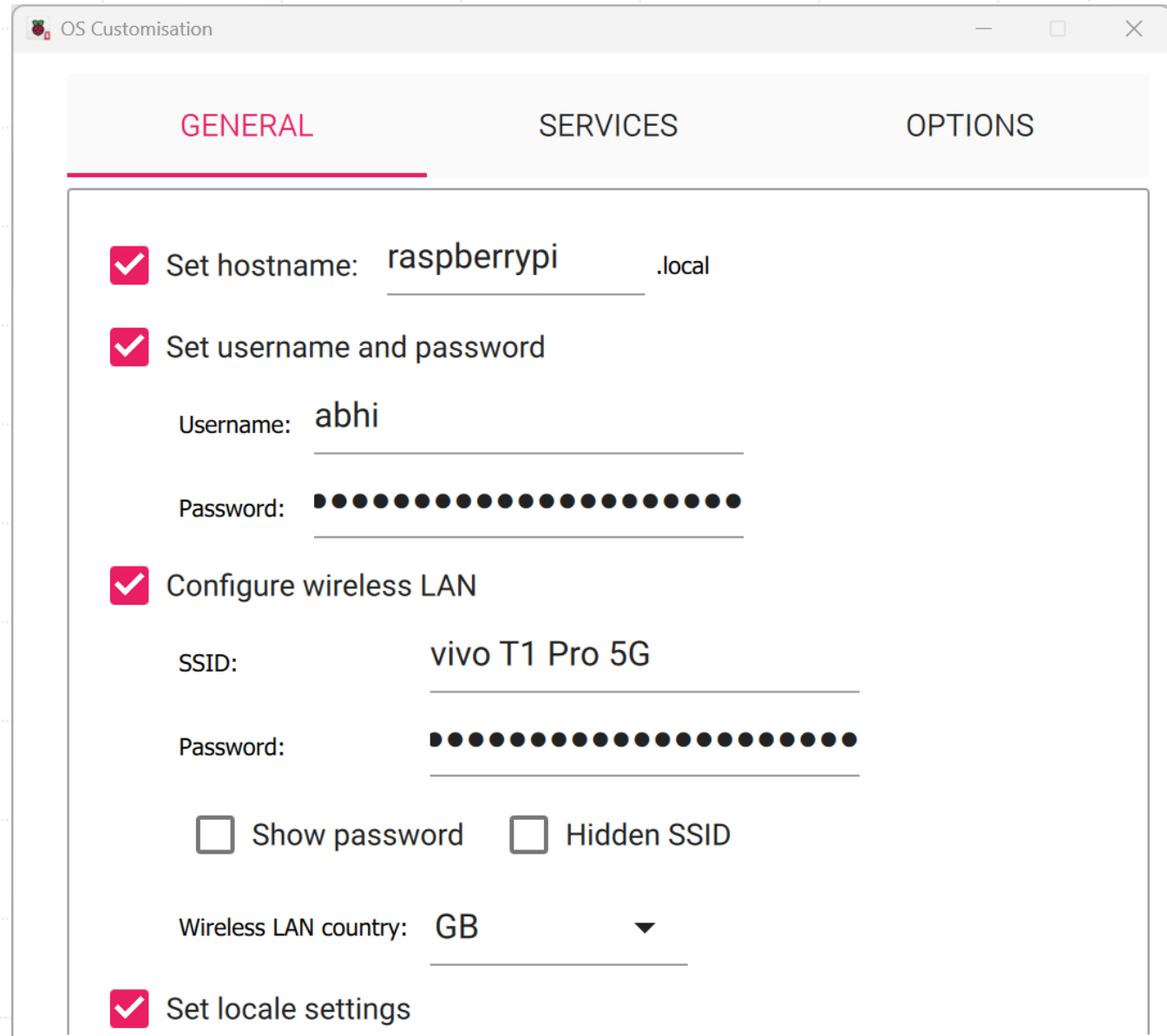
Raspberry Pi Imager

- Storage = SD Card/ USB Drive
- Next
- Edit Settings



RPi Imager

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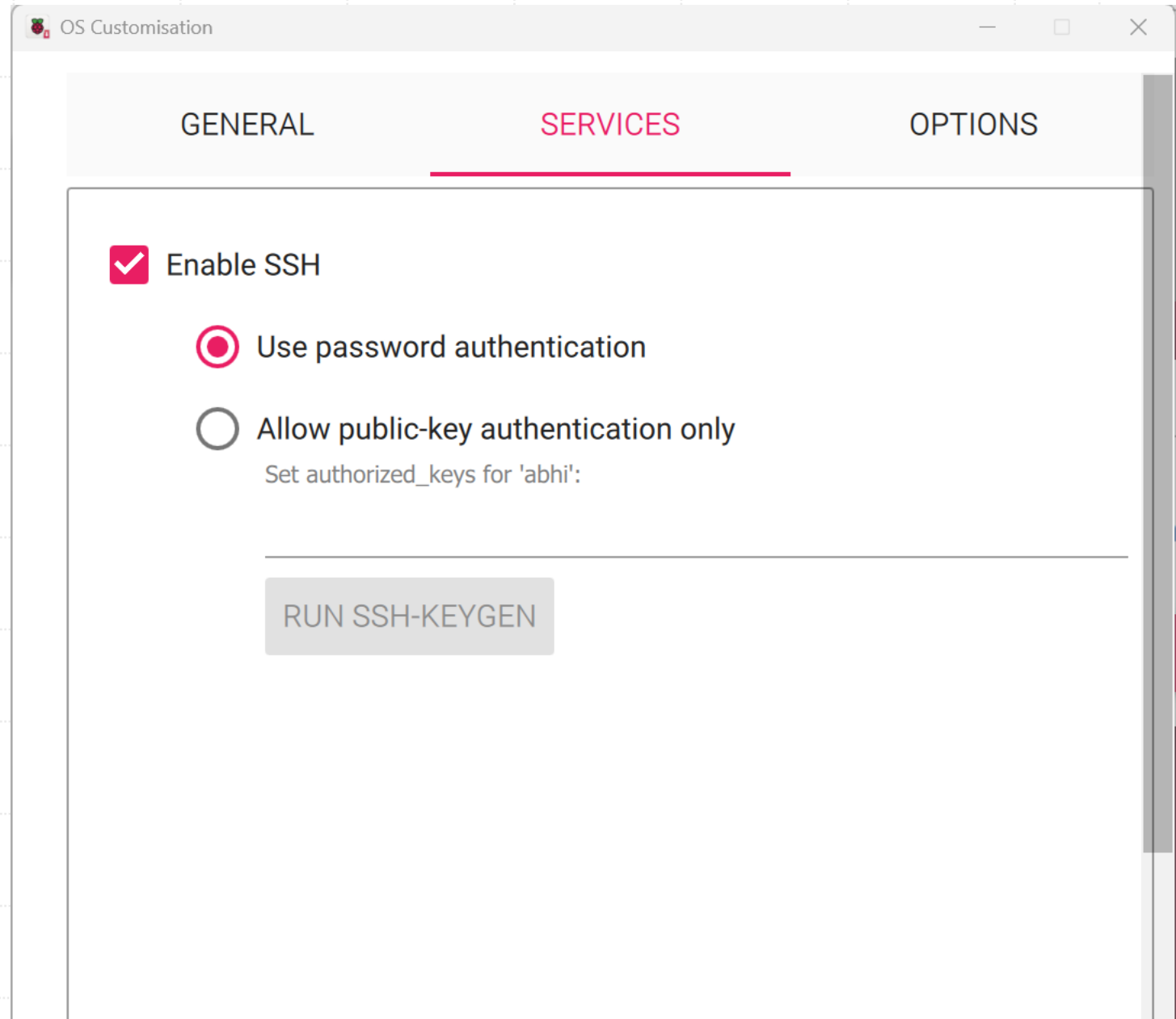


The screenshot shows the 'OS Customisation' window with the 'GENERAL' tab selected. The window has a title bar with standard Linux window controls. The 'GENERAL' tab is highlighted with a pink underline. The 'SERVICES' and 'OPTIONS' tabs are visible but not selected. The 'GENERAL' tab contains several settings, each with a checked checkbox:

- ☒ Set hostname: .local
- ☒ Set username and password
 - Username:
 - Password:
- ☒ Configure wireless LAN
 - SSID:
 - Password:
 - ☐ Show password ☐ Hidden SSID
 - Wireless LAN country:
- ☒ Set locale settings

RPi Imager

■ Services



The screenshot shows the 'OS Customisation' window with the 'SERVICES' tab selected. The 'GENERAL' and 'OPTIONS' tabs are also visible. Under the 'SERVICES' tab, the 'Enable SSH' checkbox is checked. Below it, there are two radio button options: 'Use password authentication' (selected) and 'Allow public-key authentication only'. A text field for 'Set authorized_keys for 'abhi':' is present but empty. At the bottom, there is a button labeled 'RUN SSH-KEYGEN'.

OS Customisation

GENERAL SERVICES OPTIONS

☒ Enable SSH

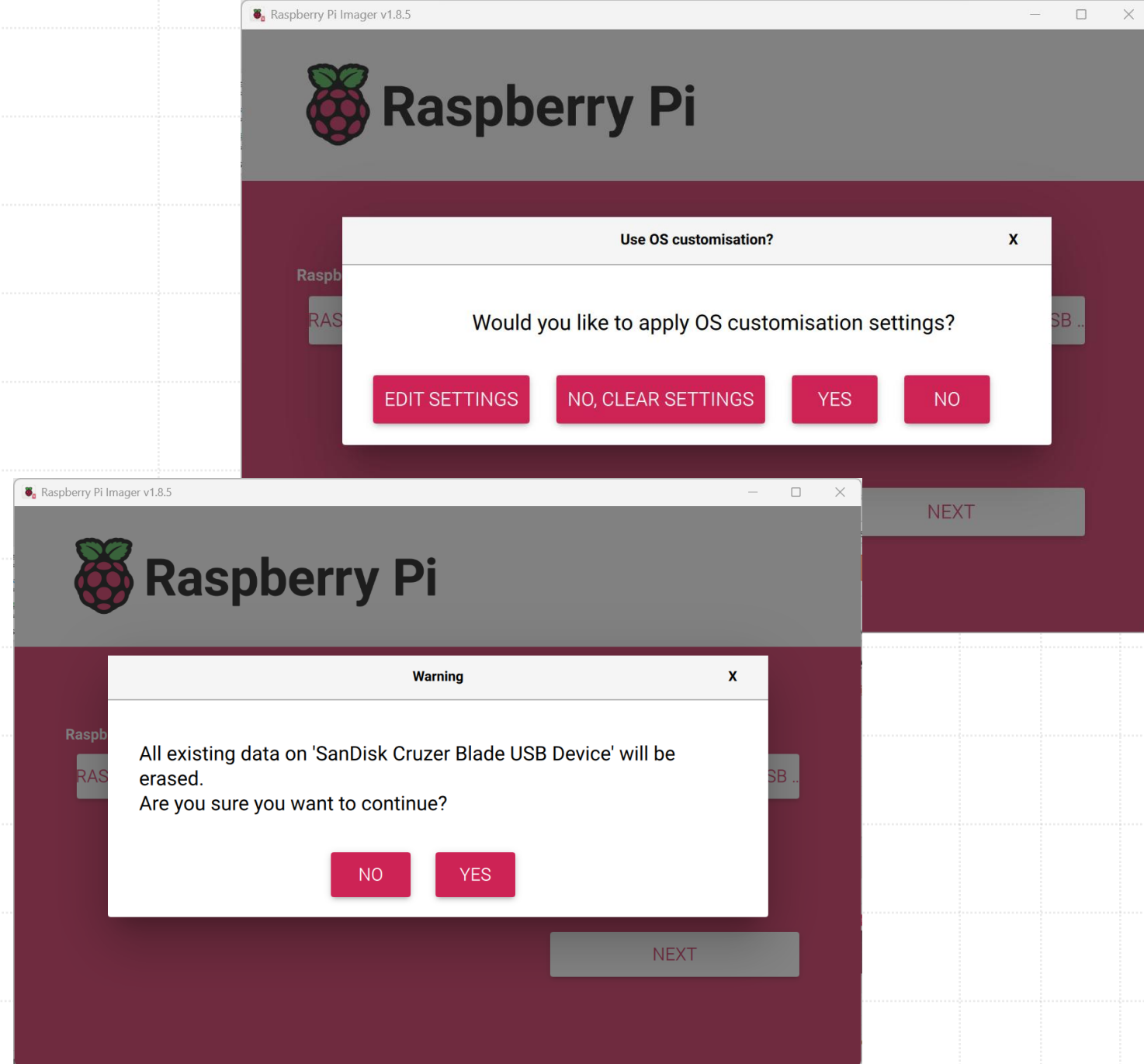
☒ Use password authentication

☐ Allow public-key authentication only
Set authorized_keys for 'abhi':

RUN SSH-KEYGEN

RPi Imager

- Apply Settings
- Format = Yes
- 10-15 Minutes



Managing SSH

- Tool – PuTTY
- To enable laptop connection with RPi



VNC Connection

- Tool – VNC Viewer
- To monitor RPi desktop on your laptop

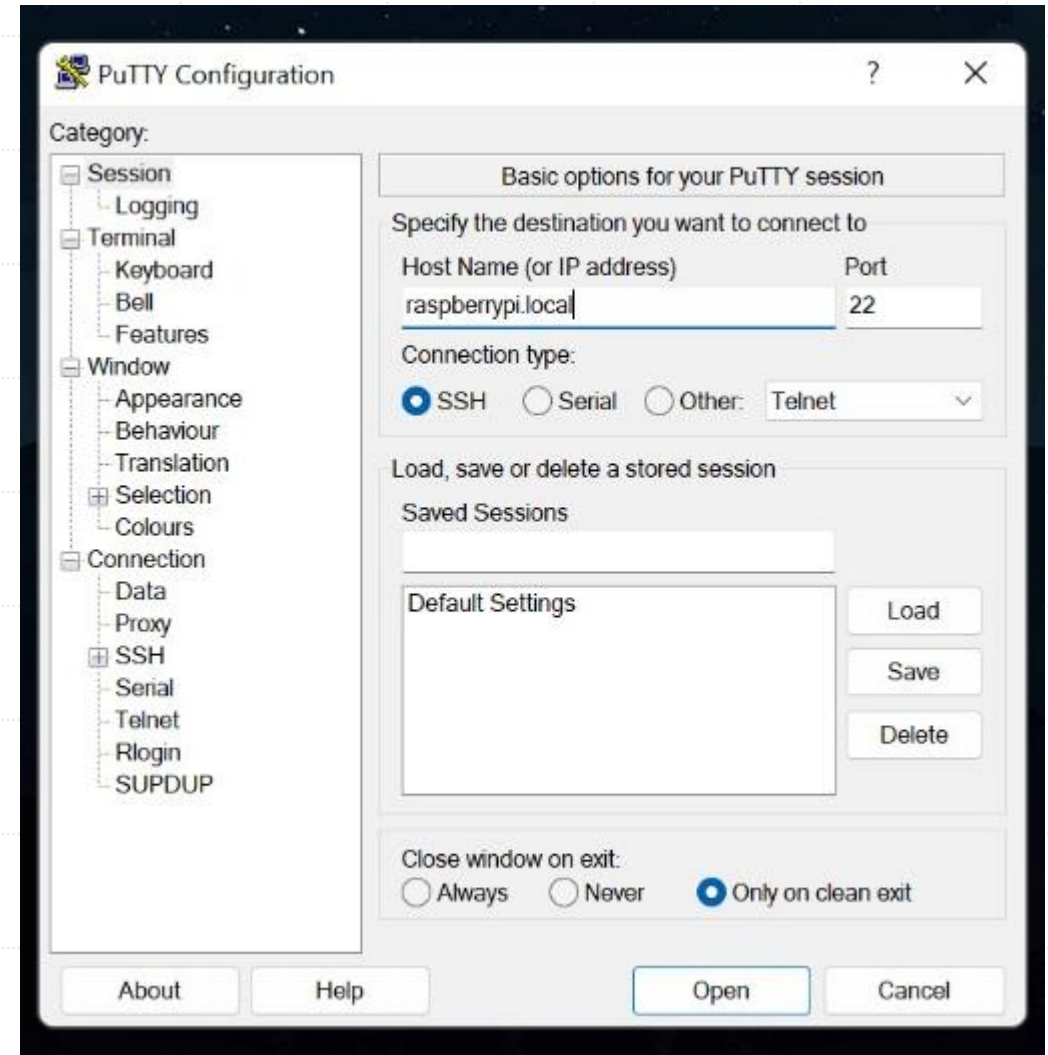


PuTTY Connection

- Ensure Laptop and RPi are connected to same network (Preferably Mobile Hotspot)
- Open PuTTY tool

PuTTY Connection

- Under "Host Name (or IP address)," type
- **raspberrypi.local**
- Port 22



PuTTY Connection

- If a PuTTY Security Alert pops up, select Accept. 6.
- In the console window, type "username" after "login as."
- Enter the password you created earlier.
- You should be able to log in to the Raspberry Pi successfully

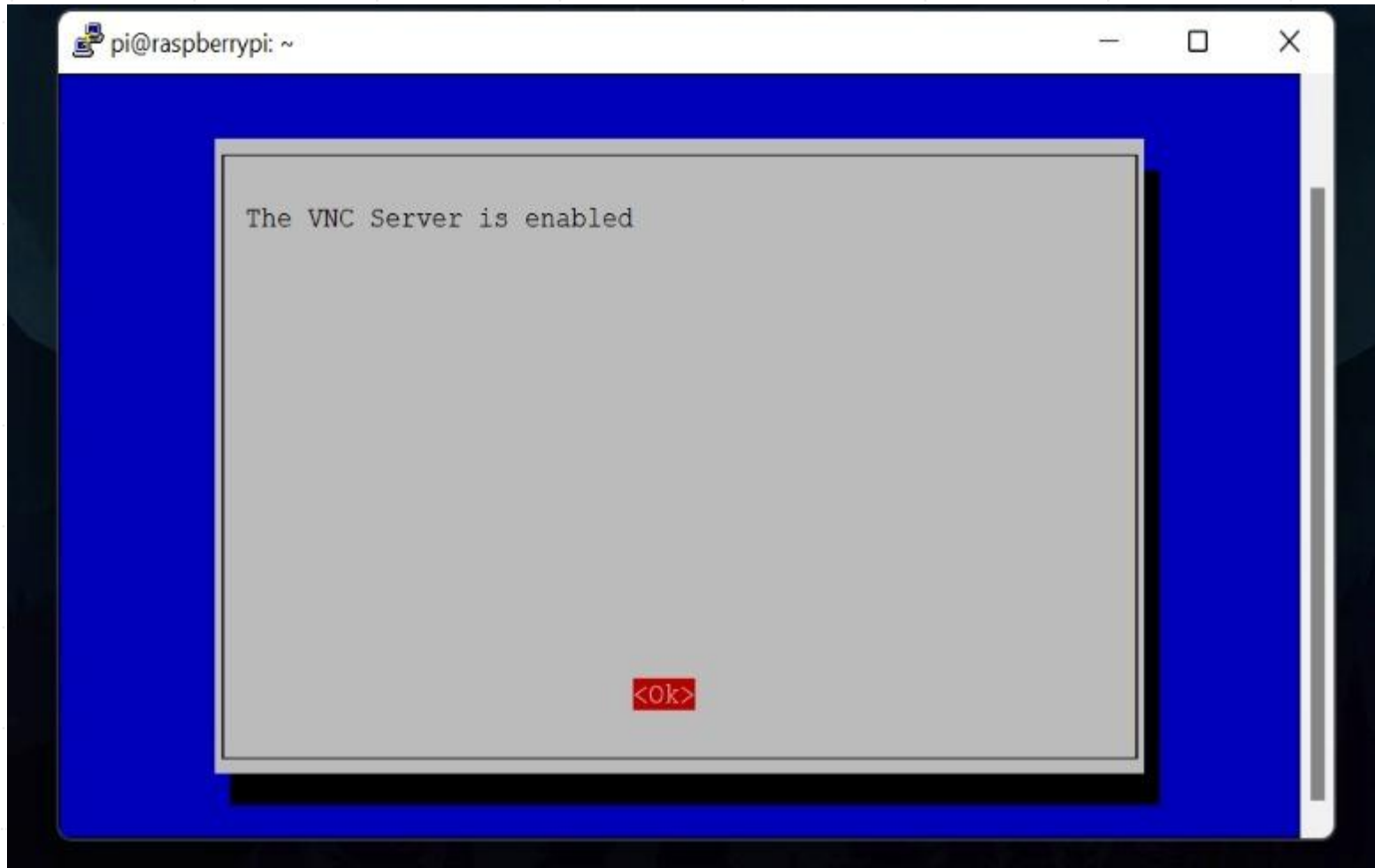
PuTTY Connection

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PuTTY Connection

- In the console window, run "sudo raspi-config" (without quotes).
- This will launch the configuration tool.
- Navigate to Interface Options using your keyboard's arrow keys and hit Enter
- Go to VNC and hit Enter.
- Select Yes to confirm that you want to enable the VNC Server.
- Hit OK to return to the homepage

PuTTY Connection



PuTTY Connection

- Navigate to System Options.
- Open "Boot/Auto Login."
- Go to Desktop Autologin and hit Enter
- This option will boot the Raspberry Pi board to its desktop environment and will automatically log you in as the pi user.
- Move to the Finish button at the bottom to apply the changes.
- Reboot the board by selecting Yes. This will disconnect you from the board

VNC Viewer

- Launch the VNC Viewer
- Click on "Use VNC Viewer without signing in."
- In the search bar, type "raspberrypi.local" (without quotes). Then, hit Enter.
- In the Identity Check pop-up, click on Continue to connect to the Raspberry Pi board.
- Enter your username (pi) and password in the respective fields. Then, click OK.